MTH-356: Linear Algebra with Application

Lecture 2 – Vectors, Span, and Basis in Machine Learning

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Contents

1 Introduction: Linear Algebra for Machine Learning

Why This Matters for Math Students

The primary objective of this course is to build a strong foundation in linear algebra tailored specifically for understanding machine learning (ML) concepts. We will prioritize visual and graphical intuition over abstract mathematical proofs to see how these operations are applied in ML. Our focus will be on answering two key questions:

- How can we visualize the operations of linear algebra?
- How are these operations relevant to solving problems in machine learning?

2 What is a Vector?

A vector can be understood from three different perspectives, each relevant in its own domain. For our purposes, we will adopt the third view.

1. **Physics View:** A quantity defined by both **magnitude** (length) and **direction**. In physics, a vector can be located anywhere in space; its starting point is not fixed.
2. **Computer Science View:** An ordered list or array of numbers. This is the fundamental way data is structured and stored in machine learning.
   - **Example:** A student's data can be represented as a vector:

     $$\text{[Student ID, Age, Weight]} \rightarrow [347, 20, 70]$$

     This is a 3-dimensional vector.
3. **Linear Algebra (for ML) View:** An arrow representing magnitude and direction that is **always rooted at the origin** of a coordinate system. This fixed-origin perspective is crucial for understanding transformations.

Vector Notation

In this course, we will represent vectors as **column vectors**. This distinguishes them from points, which are typically written as ordered pairs or tuples.

- A point at $(3, 4)$ is a location.
- A vector from the origin to that point is an object with magnitude and direction, written as:

  $$\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

3 Fundamental Vector Operations

The core of linear algebra involves "linear transformations"—systematic ways of manipulating vectors. The most basic of these are vector addition and scalar multiplication.

3.1 Vector Addition

Vector addition can be visualized as accumulating movements or "jumps" along each axis.

- **Example:** Let's add two vectors, $\vec{x}$ and $\vec{y}$.

$$\vec{x} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \quad \text{and} \quad \vec{y} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

- **Intuition:**
  1. Vector $\vec{x}$ corresponds to moving 3 units right and 4 units up.
  2. Vector $\vec{y}$ corresponds to moving 2 units right and 1 unit down.
  3. The sum $\vec{x} + \vec{y}$ is the net result of all these movements combined.
- **Calculation:**

$$\vec{x} + \vec{y} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} + \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 3+2 \\ 4+(-1) \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

3.2 Scalar Multiplication (Scaling)

Multiplying a vector by a scalar (a single number) scales its magnitude but does not change its direction (unless the scalar is negative).

- **Example:** Scale vector $\vec{y}$ by a factor of 2.

$$2 \cdot \vec{y} = 2 \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \cdot 2 \\ 2 \cdot (-1) \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \end{bmatrix}$$

- **Effects of Different Scalars:**
  - Scalar > 1: Elongates the vector (stretches it).
  - $0 <$ Scalar < 1: Shrinks the vector (squishes it).
  - Scalar < 0: Flips the vector to the opposite direction and scales it.

4 Basis Vectors and Linear Combinations

Linear Combination

The operation of scaling two or more vectors and adding them together is known as a **linear combination**. This is one of the most fundamental concepts in linear algebra. Any vector in a space can be represented as a linear combination of its basis vectors.

The standard basis vectors in 2D are two special unit vectors:

- $\hat{\imath}$ (i-hat): A unit vector along the positive x-axis. $\hat{\imath} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$
- $\hat{\jmath}$ (j-hat): A unit vector along the positive y-axis. $\hat{\jmath} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

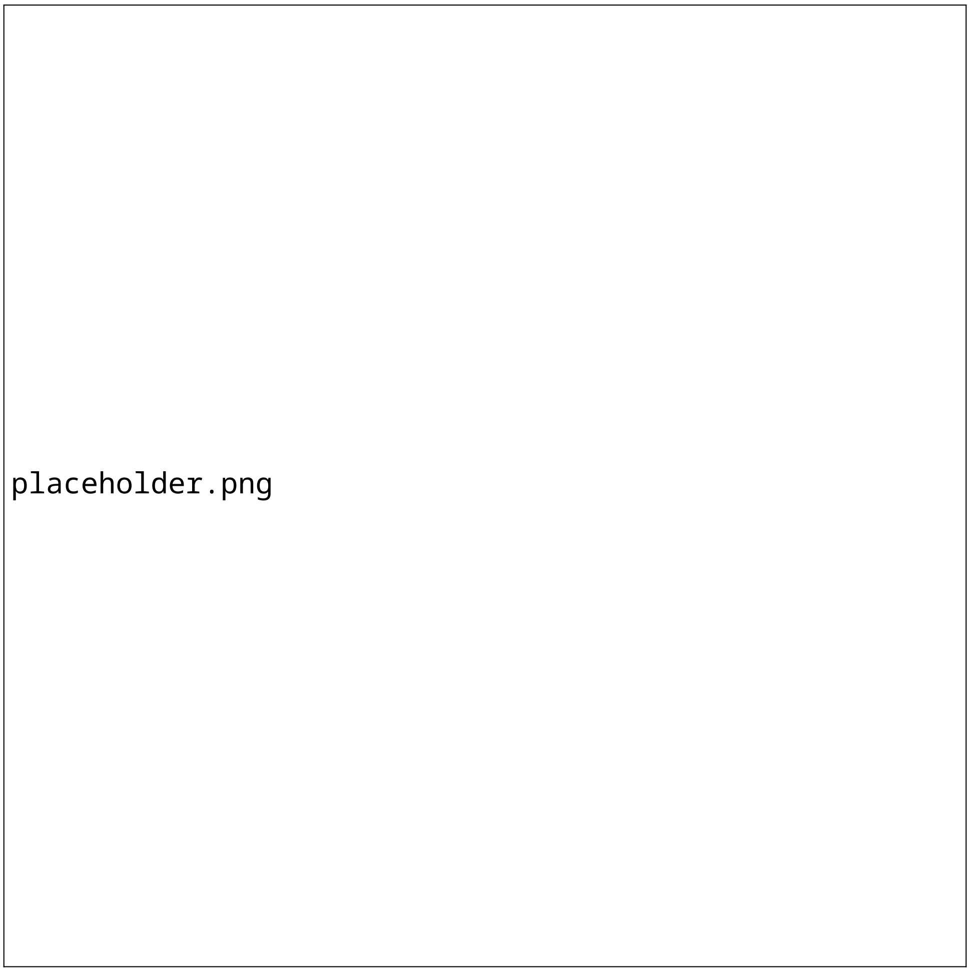

Figure 1: Visual representation of (a) Vector Addition and (b) Scalar Multiplication.

Any vector can be expressed as a linear combination of $\hat{i}$ and $\hat{j}$. For example, the vector $\vec{y} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ can be written as:

$$\vec{y} = 2\begin{bmatrix} 1 \\ 0 \end{bmatrix} + (-1)\begin{bmatrix} 0 \\ 1 \end{bmatrix} = 2\hat{i} - 1\hat{j}$$

4.1 Changing the Basis

The coordinates of a vector are relative to the basis being used. If we choose a different set of basis vectors, the same geometric vector will have different coordinates.

- Let's consider a new basis, $\vec{i_1} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\vec{j_1} = \begin{bmatrix} -0.5 \\ 1 \end{bmatrix}$.
- The same vector $\vec{y}$ can be represented as a linear combination of these new basis vectors, for instance, $\vec{y} = 1\vec{i_1} - 2\vec{j_1}$.
- This means the vector $\vec{y}$ has coordinates $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$ in the standard basis but $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$ in the $\{\vec{i_1}, \vec{j_1}\}$ basis.

This idea is critical in machine learning, where changing the basis (or "feature space") can make a problem easier to solve.

5 Span: What Can You Reach?

Span

The **span** of a set of vectors is the collection of all possible vectors that can be reached by taking linear combinations of them. It defines the "reach" or dimensionality of a set of vectors.

5.1 Span in 2D Space

- **Two Non-Parallel Vectors:** Their span is the entire 2D plane. You can reach any point.
- **Two Parallel Vectors:** Their span is only a 1D line within the 2D plane. All linear combinations remain on that line.

5.2 Span in 3D Space

- **Two Non-Parallel Vectors:** Their span is a 2D plane embedded within the 3D space.
- **Three Vectors:**
  - If the third vector lies **outside** the plane of the first two, their span is the entire 3D space.
  - If the third vector lies **within** the plane of the first two, their span is still just that 2D plane.

6 Linear Independence

Linear Independence

A set of vectors is **linearly independent** if no vector in the set can be written as a linear combination of the others. Geometrically, this means that each vector adds a **new dimension** to the span.

- **Example in 2D:**
  - A single vector $\vec{v_1}$ spans a 1D line.
  - If you add a second vector $\vec{v_2}$ that is not on that line, the span becomes a 2D plane. A new dimension was added, so $\{\vec{v_1}, \vec{v_2}\}$ are **linearly independent**.
  - If $\vec{v_2}$ lies on the same line as $\vec{v_1}$, the span remains 1D. They are **linearly dependent**.
- **Example in 3D:**
  - Two vectors $\{\vec{a}, \vec{b}\}$ span a 2D plane.
  - A third vector $\vec{c}$ is linearly independent from them if and only if it points out of that plane, expanding the span to 3D.

7 Basis (Formal Definition)

Basis

A **basis** for a vector space is a set of **linearly independent** vectors that **span** the entire space. It is the minimum set of vectors needed to "build" the entire space.

- **For a 2D plane:** A basis consists of any two linearly independent (non-parallel) vectors.
- **For a 3D space:** A basis consists of any three linearly independent vectors.

The standard basis $\{\hat{i}, \hat{j}, \hat{k}\}$ is just one of infinitely many possible bases for 3D space.

8 Relevance to Machine Learning

Machine Learning Applications

These foundational concepts are not just abstract math; they are the tools used to manipulate data in machine learning models.

- **Dimensionality Reduction:** Techniques like Principal Component Analysis (PCA) find a new basis for the data and discard the basis vectors (dimensions) that carry the least information. This is equivalent to projecting data from a high-dimensional space onto a lower-dimensional one (e.g., from a 3D space to a 2D plane).
- **Feature Engineering:** Transforming data into a higher-dimensional space can sometimes make patterns more apparent. This involves creating a new basis where a classifier (like a linear separator) can easily distinguish between classes.
- **Deep Learning:** Neural networks perform a series of linear transformations (matrix multiplications) and non-linear activations. Each layer can be seen as transforming the input data into a new vector space (a new basis) where features are more useful for the task at hand.

9 Lecture Summary

- **Vector:** For our purposes, an arrow rooted at the origin, represented by a column of numbers.
- **Linear Combination:** The fundamental operation of scaling and adding vectors.
- **Span:** The set of all vectors reachable through linear combinations. It describes the "space" covered by a set of vectors.
- **Linear Independence:** A set of vectors where each one contributes a new dimension to the span.
- **Basis:** A set of linearly independent vectors that spans the entire space, forming a coordinate system.